

# MediPilot Risk Engine & Perception System

## Technical Specification & Implementation Details

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### **Abstract**

This document serves as the comprehensive technical specification and implementation manual for the MediPilot prototype. It details the architecture of the end-to-end system, covering the Perception Layer (Track A) and the Risk Engine (Track C). MediPilot employs a strict “Fail-Closed” philosophy, utilizing deterministic clinical rule-cards, multi-vital distress floors, and conformal prediction guarantees to ensure patient safety before, during, and after machine learning interventions.

### **Contents**

# 1 System Architecture

The MediPilot system is designed around a strictly separated, parallel track architecture. This separation of concerns ensures that the machine learning models operate within tight bounds and that the core clinical safety rules can never be bypassed by unpredictable AI outputs.

## 1.1 The Fail-Closed Philosophy

In clinical artificial intelligence, a “Fail-Closed” architecture dictates that in the event of missing data, out-of-distribution inputs, or model uncertainty, the system defaults to the safest deterministic clinical rule rather than generating a potentially hallucinated or miscalibrated output.

MediPilot achieves this by placing hard clinical rules (such as red flags and multi-vital distress floors) *outside* of the model path. If the model fails or is bypassed, the rule card governs the patient’s triage band.

## 1.2 Architectural Tracks

The system is divided into three parallel tracks:

- **Track A (Perception Layer):** Owns everything between the patient/nurse’s voice and a clean, structured record. It handles speech recognition, age-aware questioning, and grammar-constrained Large Language Model (LLM) extraction.
- **Track B (Nurse Frontend):** Manages direct vital sign entry and physical kiosk interactions. (Not detailed in this document).
- **Track C (Risk Engine & Backend):** Receives structured records from Track A and vitals from Track B to orchestrate triage band assignment (Red/Yellow/Green), recheck scheduling, and hospital surge control using a combination of deterministic rules and a Gradient Boosted Decision Tree (GBDT).

# 2 Perception Layer (Track A)

The Perception Layer bridges the gap between raw human input (voice) and the strict structural requirements of the risk engine.

## 2.1 Intake & Automatic Speech Recognition (ASR)

The intake process begins with a branching gate determining if the patient is assisted or unassisted, and whether they consent to voice recording. If consent is declined, the patient undergoes an observation-only triage path without penalty.

The Automatic Speech Recognition (ASR) system utilizes a Whisper-class model fine-tuned for Indian-accented speech and code-mixed inputs (e.g., Hinglish). If the ASR confidence score drops below a safe threshold, or if an unsupported language is detected, the system immediately flags a `communication_barrier` and falls back to manual nurse text entry.

## 2.2 Age-Aware Question Tree

The patient’s age is extracted early to route them into one of three question trees:

1. **Paediatric:** Focuses on carer-reported observations, feeding, and behaviour.
2. **Adult/Adolescent:** Focuses on chief complaint, pain severity (0-10), and pregnancy status.

- 3. **Geriatric:** Focuses on baseline functional decline, confusion, and polypharmacy.

## 2.3 Grammar-Constrained LLM Structurer

A Large Language Model is employed to extract observations from the transcript. However, it operates under strict grammar constraints and JSON schema validation.

- **Allowed Fields:** `chief_complaint`, `onset_minutes`, `self_reported_severity`, `symptoms`, `medications`, `pregnancy_status`.
- **Prohibited Operations:** The LLM has *no production rule* for generating a diagnosis or acuity band. The grammar rejects any token generation outside the allowed schema.

## 2.4 Reliability Flags

Track A assembles observability signals as pure boolean flags (e.g., `non_assisted`, `analgesia_given`, `health_literacy_low`). It does not discount or weight these signals; it strictly reports them to Track C, which handles asymmetric reliability discounting.

# 3 Hard Clinical Safety Guarantees

Before the machine learning risk engine evaluates a patient, the system applies a series of deterministic rule-cards. These rules provide clinical floors that the model cannot override.

## 3.1 Deterministic Red-Flags

Extracted observations (e.g., `altered_consciousness`) are matched against a static YAML configuration table. If a match occurs, the patient is immediately assigned a Red band. The ML risk engine is completely bypassed, as these presentations are blatantly life-threatening.

## 3.2 Abnormal Vital Floors (Stratum-Specific)

Different age strata have distinct physiological norms. The system enforces an `abnormal_vital_floor` that dictates, for example, that a febrile toddler or a geriatric patient with any abnormal vital sign cannot score below a Yellow band, regardless of the model's confidence.

## 3.3 Multi-Vital Distress Floor

Patients presenting with  $\geq 4$  simultaneously abnormal vital signs meet Systemic Inflammatory Response Syndrome (SIRS) or shock criteria by definition. Since the model is trained on data where such extreme, single-snapshot presentations are sparse, it may underestimate the risk. The rule card overrides this by forcing a minimum Yellow band.

## 3.4 Critical Derangement Floor (New)

If a patient presents with a single vital sign that is deeply deranged—indicating peri-arrest physiology—the system forces an immediate Red band. The critical thresholds across all strata include:

- **GCS**  $\leq 8$ : Unresponsive / airway at risk.
- **SpO2**  $< 85\%$ : Severe hypoxaemia.
- **Systolic BP**  $< 70$  mmHg: Decompensated shock.

- $\geq 6$  **Abnormal Vitals**: Near-total systemic derangement.

These values represent individually life-threatening states rather than a composite score. The model probability  $p_{\text{model}}$  is clamped to exceed the red threshold ( $p \geq p_{\text{red}}^*$ ) unconditionally.

## 4 Machine Learning Pipeline (Track C)

When deterministic rules do not fire, the patient’s record is evaluated by the Machine Learning risk engine. This engine predicts the probability of a critical outcome using a Gradient Boosted Decision Tree (GBDT) pipeline.

### 4.1 Synthetic Data Generation

To ensure robust training, a highly realistic synthetic dataset of 20,000 patient records is generated. The generator uses latent health and frailty models combined with Bernoulli outcome sampling to simulate real-world emergency department distributions. The training labels represent clinical outcomes, not heuristic triage bands.

### 4.2 Feature Registry and Leakage Prevention

The `HistGradientBoostingClassifier` relies on a strict feature registry that bans data leakage. Features such as `condition_id` or `acuity_band` are explicitly prohibited. If an unregistered or banned feature attempts to enter the model, the system fails closed.

### 4.3 Calibration and Conformal Prediction

Raw model scores are often miscalibrated. MediPilot applies Isotonic Regression independently per age stratum to ensure the output probabilities represent true risk.

Furthermore, Mondrian Split-Conformal Prediction is utilized to provide strict statistical guarantees on uncertainty. Instead of a single point prediction, the model outputs a calibrated probability alongside an uncertainty band, calculated uniquely for each stratum (e.g., infants have different uncertainty bounds than adults).

### 4.4 Cost-Ratio ( $R$ ) Sweep

The conversion of a continuous risk probability into a discrete triage band (Green, Yellow, Red) is governed by a dynamic cost-ratio,  $R$ . The threshold  $p^*$  is derived using the Bayes-optimal formula:

$$p^* = \frac{1}{1 + R}$$

As a hospital’s risk tolerance decreases (higher  $R$ ), the decision boundary shifts downwards, making it easier for a patient to be assigned a higher acuity band. The model’s objective probability remains unchanged, but the policy threshold dynamically adapts to the hospital’s current capacity and risk profile.

## 5 Backend Orchestration & Operations

The final layer of the architecture acts upon the decisions generated by both the clinical rules and the risk engine.

## 5.1 Band Engine

The Band Engine is responsible for consolidating the outputs from the deterministic rule-cards (e.g., Red-Flags, safety floors) and the GBDT model. It determines the patient’s final acuity band (Red, Yellow, or Green) based on a strict precedence hierarchy, ensuring that model probabilities never override hard clinical safety boundaries.

## 5.2 Surge Controller

In high-stress emergency department environments, bed capacity can be overwhelmed. The Surge Controller monitors overall hospital capacity and dynamically applies surge policies. During severe overload, it can restrict the assignment of the Red band, capping the number of patients permitted in the critical queue and forcing non-critical patients downward, ensuring that only the highest acuity cases receive immediate attention.

## 5.3 Recheck Scheduler

Patients placed in lower acuity bands or waiting in queues require continuous monitoring. The Recheck Scheduler uses age-stratified cadence matrices (configured in `band_cadence.yaml`) to automatically calculate when a nurse must manually remeasure vital signs. The cadence logic adjusts dynamically based on the staleness of the current vitals and the patient’s assigned band.

# 6 Validation & Evaluation Gates

To ensure clinical and mathematical integrity, MediPilot implements strict honesty gates and a comprehensive invariant testing suite.

## 6.1 Invariant Testing

The test suite consists of 90+ tests spanning unit logic, conformal prediction behavior, and clinical rules. The tests ensure that core invariants—such as missing value disciplines (NaN vs. sentinels) and fail-closed behaviors—are preserved under all conditions.

## 6.2 Honesty Gates

During model evaluation, the build process must pass multiple rigid gates to deploy an artifact:

- **Prevalence Gate:** The dataset prevalence must fall within an expected clinical range ([5%, 20%]).
- **AUPRC Lift:** The model’s Area Under the Precision-Recall Curve (AUPRC) must beat the baseline prevalence by at least  $1.5\times$ .
- **AUROC Canary:** Because the synthetic dataset generates labels with injected frailty and Bernoulli noise, a Bayes-optimal model will never achieve perfect prediction. An AUROC exceeding 0.95 triggers a failure, flagging potential data leakage (e.g., the model learning a diagnosis rather than outcome risk).
- **Conformal Coverage:** The Mondrian conformal calibration must guarantee a user-defined coverage threshold (typically  $\geq 90\%$ ) for true outcome inclusion across all age strata.

These gates prevent overtuning, leakage, and unsafe deployment of non-robust machine learning artifacts.